



A Knowledge-Based Model for Constructability Assessment of Buildings Design Using BIM

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Abstract. Economic and time efficiency can be attained in the construction industry by applying the principles of constructability. Existing empirical studies demonstrate that incorporating these principles into initial stages of design maximise outcomes for all stakeholders including designers, contractors, and clients. Considering the complexity of current building design processes, there is a need to provide a decision support tool that can help designers in designing for constructability based on embedded information within the design model. Such a tool would be most beneficial at the conceptual design stage so that constructability is factored into the design solution starting from its inception. Therefore, this research investigates how contemporary process- and object-oriented models can be used to provide a mechanism that represents the subjectivity of design constructability to inform decision making. Consequently, it proposes a BIM-based model using embedded information within the design environment to conduct the assessment. The modelling framework is composed of three key parts: The *Constructability Model (CM)* which formulates user-based knowledge; the *BIM Design Model* which provides required data for the assessment; and the *Assessment Model (AM)* which reasons with the formulated knowledge and the BIM Design Model. The modelling framework is implemented in C#, using .NET Frameworks and Revit API. This paper demonstrates that using this framework, constructability related information can be captured and reasoned with to inform decisions at the early stages of the design process.

Keywords: Constructability assessment · Building information modelling · Building design

1 Introduction

The constructability concept aims to integrate engineering, construction and operation knowledge and experience to better achieve project objectives [1]. The term is defined by the Construction Industry Institute [2] as “the optimum use of construction knowledge and experience in planning, design, procurement, and field operations to achieve overall project objectives”. Similarly, the Construction Industry Research and Information Association (CIRIA) defines the buildability as “the extent to which the

design of a building facilitates ease of construction, subject to overall requirements for the completed building” [3].

The importance of deploying the constructability concept at the early design stage stems from the criticality of this phase in any architectural, engineering and construction (AEC) industry project. The building design is a fundamentally complex process, with the most influential design decisions being made in the early design stage [4]. This includes the consideration of design constructability, which is often ignored by designers and building clients until the commencement of the construction phase, when they are confronted by potential adverse and costly realities [5].

The significance of designing for constructability is globally recognised in the construction industry [6]. To date, several studies attempted to address the subject and accommodate its controversy aspects [7]. They adopted different approaches to benchmark design constructability and to enable the objective evaluation of abstract concepts. As a result, various techniques and methods have been developed to improve design constructability, including developing guidelines, checklists, expert systems, and empirical formulas [8–11]. However, barriers to implementing constructability are present in design practice. This is evidenced by the significant efforts, time, and human resources required to implement the concept within the design environment, which discourages many practitioners from considering constructability in their designs [12].

Researchers have investigated the employment of advanced ICT capabilities within the architecture, engineering, and construction (AEC) industry to address gaps. Building information modelling (BIM) is the most powerful technique available to effectively implement information modelling. BIM design tools offer great capabilities in managing vast amounts of information embedded in the building model, from initiation to demolition. This has enabled its adoption for assessing aspects such as cost, energy, functionality, aesthetics, and constructability [13]. In addition, the capability of implementing parametric design rules associated with building elements allows for dynamic changes during design development to explore various alternatives [14]. However, the exploitation of such technologies for implementing constructability has not been fully realised [15]. The study, therefore, investigates design-stage assessment of design constructability by examining contemporary process and object-oriented models.

This paper presents an original assessment framework to measure constructability of BIM-based design solutions. The paper discusses incentives to design for constructability, and the possible application of BIM technologies to facilitate such a process. The sections in this paper describe the proposed modelling framework and its components, and how they contribute to the overall assessment mechanism when modelling design constructability. An overview of the framework implementation in a prototype is also included in the form of a plug-in to BIM software Revit, to operate on its created design models.

2 Literature Review

2.1 Benefits of Designing for Constructability

The introduction of constructability principles in the construction industry has well-recognised benefits for owners, contractors and designers [2, 6]. While some of these benefits are tangible and manifest in the form of cost, time, quality and safety etc., others are more subjective in their nature and are observed in the sense of their physiological and psychological rewards for the project stakeholders, including client satisfaction [16].

With respect to the cost objective, identified benefits by previous studies included aspects such as savings in the total project cost [17–19]. This is estimated by 1–14% of capital cost [20]. Others expressed it as an increase in the cost-effectiveness [21], resulting in a lower cost of bidding [22], reduced site labour [23], and better usage of resources [24]. In terms of time, implementation of the concept improves the project productivity [21, 25] and reduces outage duration [24] for an early completion of the construction phase [16, 18, 19, 21, 24]. In respect of the project quality, application of constructability principles facilitates the delivery of higher quality of built products [9, 18, 21, 24]. With regards to safety, consideration of constructability results in a safer on-site environment [24, 26, 27].

However, those benefits may be extended further to cover the entire building process, including aspects such as: improving planning perception, materials acquisition, design solutions, construction approaches, site management, teamwork, job satisfaction, project performance, and stakeholder involvement and satisfaction [16].

2.2 Associated Challenges with Constructability Assessment

Constructability encounters many challenges in practical implementation [12]. One of the main obstacles is knowledge acquisition and representation to benchmark the extent of constructability principles application in design solutions [8]. This is mainly due to the nature of construction knowledge and its subjectivity from one constructor to another, given contrasting construction capabilities and needs. One way around this obstacle is to employ survey and interview approaches to extract human elements of construction knowledge and experience [7, 11, 15, 23, 28–34]. However, studies that adopt such approaches are only valid on the particular place of the study where the knowledge was originally collected [35]. Furthermore, the formulated knowledge is not captured from a user perspective and hence will not reflect their construction capabilities and preferences. Consequently, this would undermine the accuracy of obtained assessment results that are meant to advise on a constructor ability to build a design [36].

Another big challenge to design for constructability is the lack of a design tool to effectively apply the captured knowledge and map it onto design features. Current methods demand laborious efforts and resources to execute assessment calculations and interpret their outcomes. The dynamic design process and the need for ongoing modifications in designed products necessitate automation of the constructability assessment routine whenever design changes are introduced [37].

2.3 BIM for Constructability Assessment

BIM presents a substantial opportunity to improve design constructability using its rich information repository. It enhances the integration between the design and construction processes, enabling improved quality with savings in project cost and time [38]. Object-oriented models are capable of quantifying constructability, whereby designers can observe the effects of design decisions [15].

In addition, BIM-enabled tools can electronically model and manage the vast amount of a building's information throughout its entire lifecycle. Such information can be used to estimate, schedule, detail, automate fabrication drawing, and plan construction activities. Furthermore, incorporating the time dimension in the model allows different design and execution plans to be explored and tested for better design constructability [33].

Realising the benefits of BIM-enabled constructability assessment, many studies explored the subject for the full adoption in the construction industry [15, 33, 39–45]. However, the ASCE Constructability and Construction Research Council stated in one of its special journals that “*The potential of new technology-based tools such as 4D CAD or BIM have not been fully realized. This area could also include validation of new constructability software tools*” [46].

3 Current Limitations and Emerging Issues

Despite the recognition of constructability benefits and its potential to facilitate the construction process and meet set objectives, its implementation in a method or tool still stands as a challenge. In modern practice, evaluating design constructability paradigms is a complex process and demands more efforts, resources, and time than can usually be devoted to it in real construction projects. The design team has limited technical support to oversee and assess the possible consequences of decisions taken at various stages with respect to constructability during the design stage. Many constructability aspects are left out of consideration in favour of at a later stage, when it is too late to improve the design constructability performance. There is a much greater need to enhance and support the process using specialised tools during the conceptual design stage, where critical decisions are made, rather than during the later detailing stages where changes are more complex and costly [47, 48].

No current tools provide the necessary construction knowledge to inform design decisions based on constructability considerations. To date, research has tended to focus on formulating guidelines and measures for designers to follow rather than developing a mechanism or tool to support their application during the design process. The lack of a decision-support tool that quantifies design constructability is identified as the major cause of poor constructability performance in most projects [8].

BIM technologies have emerged as potential platforms for facilitating the design process of buildings. However, the potential use of their capabilities to design for constructability has not been fully realised [15]. Therefore, this research attempts to address the question of how to map and model design constructability with the

employment of knowledge-based systems and data modelling techniques to inform design decisions.

4 A Proposed BIM-Based Constructability Assessment Framework

The proposed assessment mechanism separates between the formulation of construction knowledge and carrying out the assessment processes. The modelling framework is composed of three key parts: the *Constructability Model (CM)* which formulates user-based knowledge; the *BIM Design Model* which provides required data for the assessment; and the *Assessment Model (AM)* which reasons the formulated knowledge onto design features, as illustrated in Fig. 1.

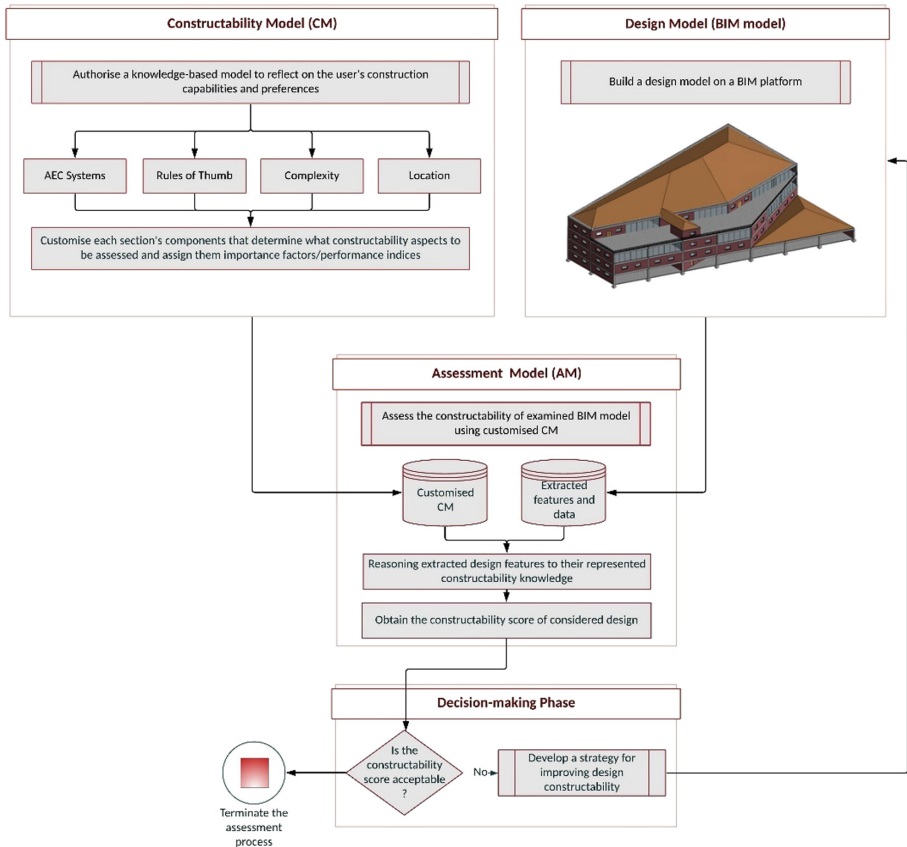


Fig. 1. Components of the proposed modelling framework

4.1 The BIM Design Model

The BIM design model refers to the digital building model that needs to be assessed for constructability. At this stage, designers build their design model using BIM authoring tool to a suitable level of details (LOD). Such information is an essential input for the assessment process, and consequently any delivered outcomes. Therefore, BIM-enabled tools should allow for the extraction of features extraction from their product models for the purpose of constructability analysis.

4.2 The Constructability Model (CM)

The CM is the knowledge-based model used for benchmarking the constructability of design solutions. Its information representations seek to model constructors' capabilities in terms of what they can build, and defining their preferences using various construction systems and methods. The represented knowledge is captured directly from users to impose their design objectives and meet particular project requirements. This is typically by enabling them to customise their own CMs. A specialised CM would typically be authored once for every type of project (e.g. multi-storey office buildings, multi-story car parks, and residential buildings, etc.), and is re-used many times for similar project types.

Figure 2 below illustrates the proposed CM. It consists of four main components: AEC Systems, Rules of Thumb, Complexity, and Location. The model components are designed to accommodate both quantitative and qualitative assessment of the design constructability, making ultimate use of embedded BIM data to model constructability in its multidimensional aspects. The contents of these components, extracted from users at the customisation stage, will guide the AM at the assessment stage in interpreting associated information with the BIM model and inform decision making.

CM: AEC Systems

The AEC systems module assesses the constructability of selected design elements based on available resources and imposed design constraints. With BIM-enabled, accurate estimation of design quantities, materials, and specifications, it is now possible to approximate demand for construction resources based on a scheduled program. This can be matched against accessible resources for design-builders to evaluate if they are capable to build it. Such evaluation of demand for resources with the available supply can be automated, having the first part accessible within the BIM model while building a resource database for the second part, which reflects users' construction resources. Consequently, the lack of a specific type of resources could be flagged at a specific stage of construction for the entire project period. Scores and indices may be assigned reflecting the intensity of observed shortage of resources. This is achieved by identifying the constructability factors and attributes to be considered during the assessment process based on the user's needs, and then calculating constructability indices of different design elements using analytical hierarchy process (AHP) method [49]. It ranks the elements based on their constructability from users' perspectives and hence enabling them to decide between alternative designs.

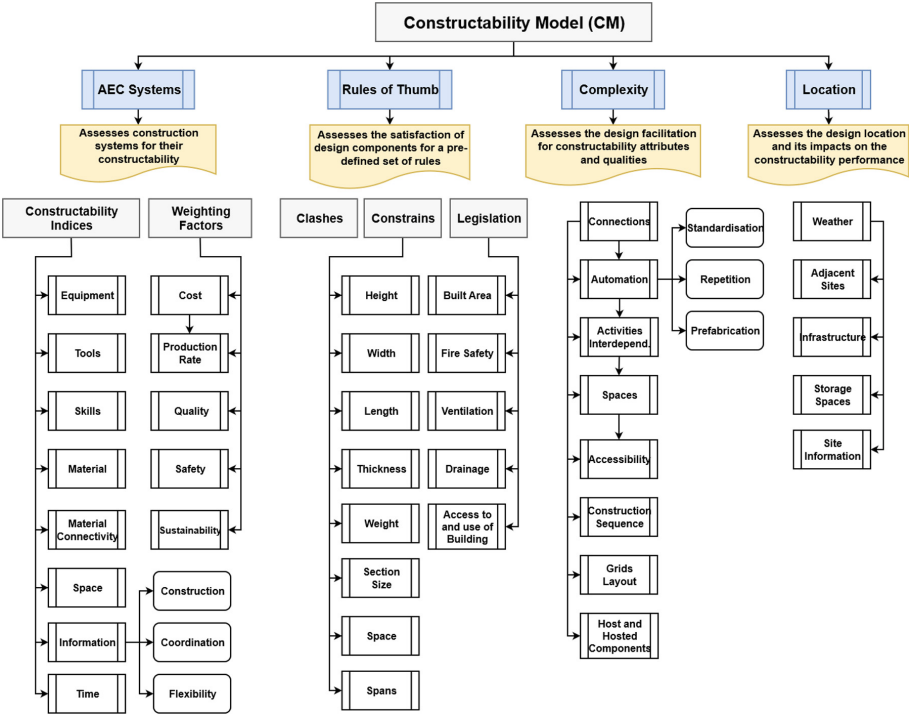


Fig. 2. Proposed CM hierarchy

CM: Rules of Thumb

This feature of the model allows users to assign a set of rules that need to be satisfied in their considered design. It takes advantage of a rule-based system to assess the design constructability based on available information in the design platform. These rules are applied to impose the design limitations and constraints in terms of spacing, layout or dimensions, which may later affect the construction process.

When customising this part of the assessment model, it is possible for users to enable rules that impose design constraints (restrictions of weight, height, length, width, etc.). Such restrictions could be applied based on the availability of elements, mode of transportation, site accessibility, available storage space, methods of constructions/installation and available working space. The rules to apply for a specific design can be activated when form the CM (not all rules need to be applied for all designs) to suit the given conditions. Although users could always opt to extend the package to include more rules, this might require some programming skills.

During the assessment process, the AM verifies the compliance of BIM model’s components with enabled rules, assigns them weights (as specified by the user in the CM) and then determines a final score representing the constructability index based on these rules. By adding the design rules feature, the proposed framework is the first of its type to combine a numerical assessment system and a rule-based system, allowing for both quantitative and qualitative approaches when modelling design constructability.

CM: Complexity

The complexity module assesses the effects of selected design solutions in facilitating various constructability aspects during the construction process, such as the intricacy of the design, automation of the process and uncertainty associated with its different aspects. The CM introduces this feature to exploit BIM data in producing performance indicators that capture the existence of such qualities in a design solution. Features such as visualisation and interaction with the BIM model within a 3D environment facilitated by nowadays design tools, enable the users to better assess their designs interactively. Detailed construction simulations or simpler 4D animations also support in the decision-making process of the designer to improve constructability.

CM: Location

The location module assesses the influence of surrounding environment factors on the construction process and how that is factored in the considered design. Aspects such as weather in the region and site conditions should be catered for in selected design elements and the way they are installed. Additionally, site accessibility and proximity to delivery sources play a vital role in choosing construction methods (e.g., precast or *in situ* casting for concrete components). In the proposed CM, the assessment of these components is based on available information within the BIM model that can be employed for this part, with some user inputs. The high-level assessment of these aspects has been discussed in [37].

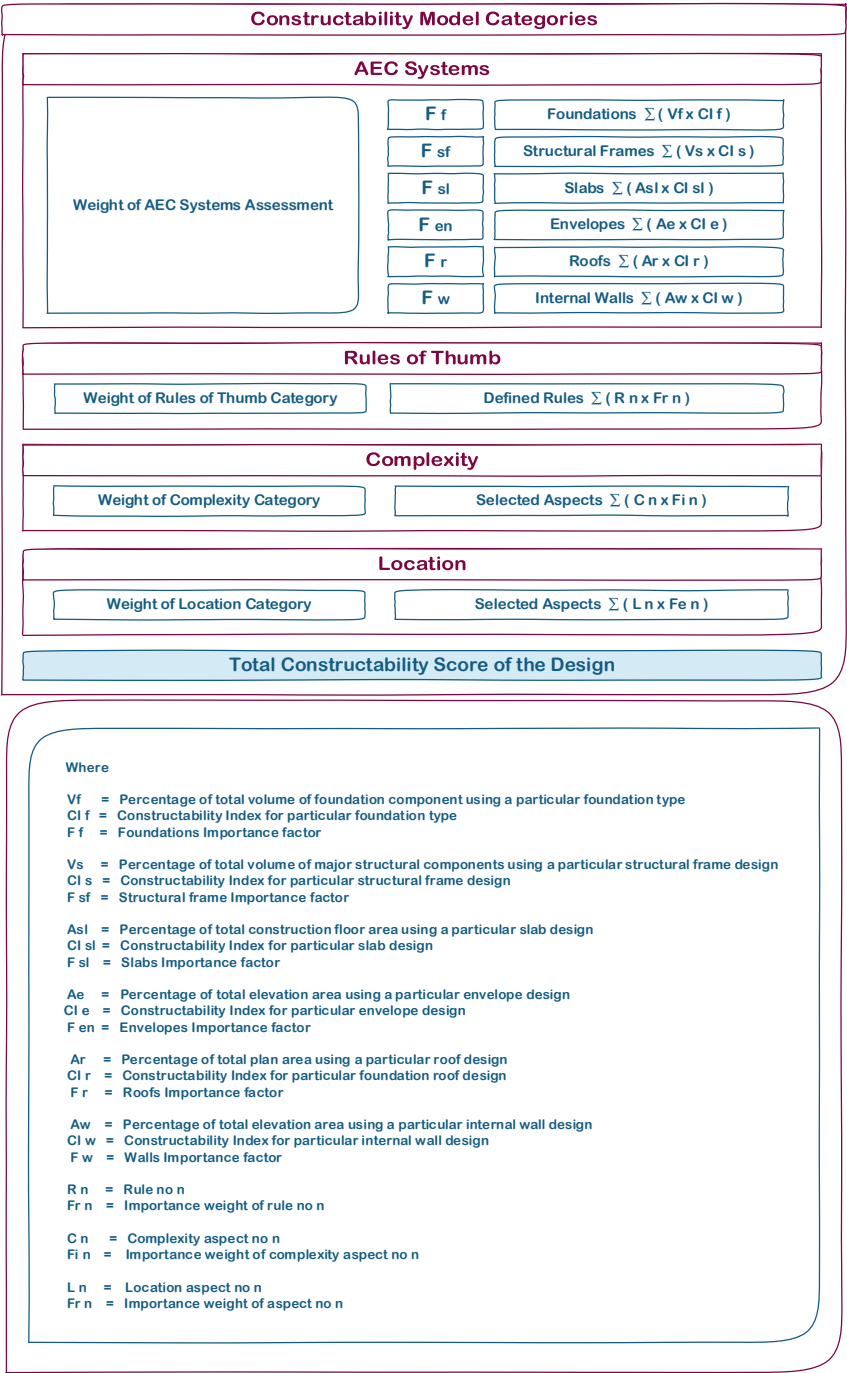
4.3 The Assessment Model (AM)

This process maps the customised CM on the actual BIM design model to benchmark its constructability. The design model is assessed based on selected AEC Systems and their suitability for its potential constructor's capabilities, satisfaction with enabled rules, complexity, and considerations of the project location. The AM extracts necessary information from the BIM model to process configured sections within the CM. The mechanism for calculating the Constructability score using the introduced model is as illustrated in Fig. 3.

5 Implementation of Constructability Assessment Prototype

The model was implemented in a prototype using object-oriented programming in a C# application. The prototype was developed using .NET Framework as a plug-in to BIM software Revit in order to operate on its created design models. The prototype was tested using typical design case studies, which has proved its usefulness in informing constructability decision-making. It also enabled the exploration and evaluation of what-if scenarios in design iterations, and construction methods.

Prototype development is based upon the elicited use-case to guide the programming direction, as demonstrated in Fig. 4. It shows the prototype functioning in four parts, namely: customising a new CM; modifying the customised CM for another use; interacting with the developed BIM model (initial analysis for its quantities), and assessing the design constructability (running the AM based on a selected CM).



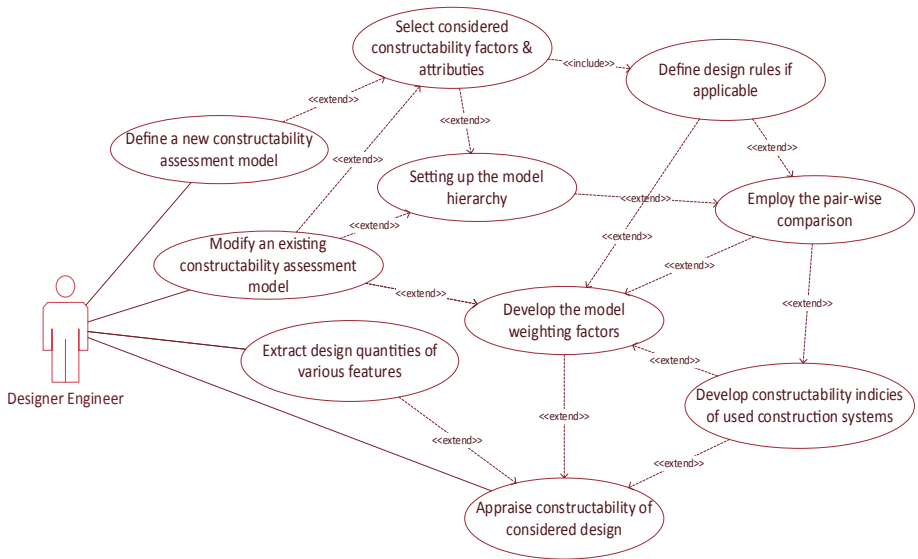


Fig. 4. Use case

The implemented prototype allows users to explore different design alternatives and decide on a design based on its constructability performance. Such a feature will enable the design optimisation and observe the difference due to using different construction systems as Fig. 5 shows.

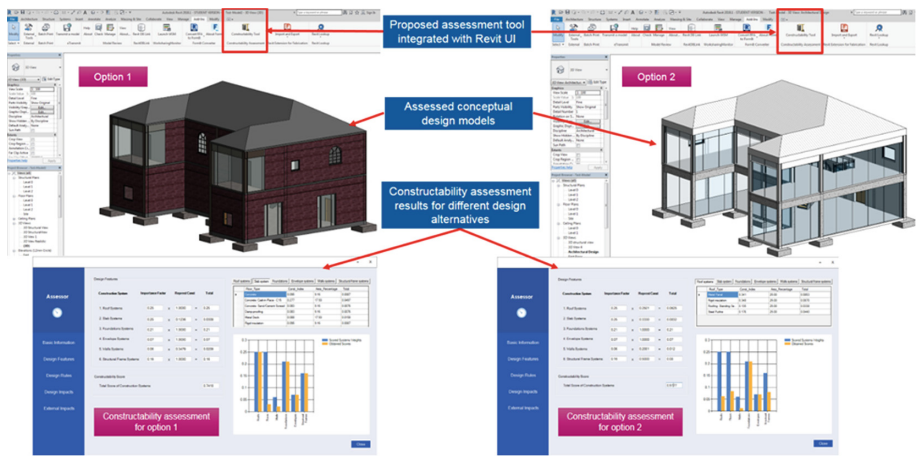


Fig. 5. Constructability assessment of design alternatives

6 Conclusion

Despite the recognised benefits of designing for constructability, it has been challenging to devise a tool that implements the concept. The paper identified the need for a decision-support tool that measures design constructability, as the major cause of poor constructability performance in most projects. Consequently, it investigated design-stage assessment of design constructability by examining contemporary process and object-oriented models.

The paper presented a framework to capture constructability related information while accommodating requirements of contrasting constructors' profiles. The framework is designed to exploit benefits of construction knowledge-based systems, object-based programming technology, and decision-making tools for modelling design constructability. It demonstrated the utilisation of BIM in constructability assessment of buildings design options through feature mapping and modelling technology.

This paper concludes that the devised modelling framework can be used to represent the subjectivity of constructability in its multidimensional aspects and inform decisions at the early stages. It enables the exploration of design alternatives for their constructability performances and hence an improvement to meet the design objectives.

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